CompSci 469/469G
Intro to Computer Security
Format String Attack - Activity

Have you formed a group?

- If you do not have a group yet, Talk to your peers and form a group of 2 persons
- Find your group no: Go to canvas -> people

Overview

Task 1: Writing to memory with %n and Examining Memory with gdb

Task 2: Writing to memory with %hn and Examining Memory with gdb

Task 3 [optional]: Writing to memory with %hhn and Examining Memory with gdb

Task 1: %n

- **Step 1:** Download string_format_write2.c from the demo module on Canvas
- **Step 2:** Compile the program:

  gcc -g -fno-omit-frame-pointer -O0 format_string_write2.c -o vul

- **Step 3:** Run the executable file first and observe the output:

  ./vul

Task 1: %n

- **Step 4:** Let's debug the program:

  gdb ./vul

- **Step 5:** In the interactive gdb shell enter this commands:

  break main

  run

  next [continue entering next until you see the line printf("abcd%n", &i);)

  .

  .

  p &i

  x/16xb &i

At this point, observe the memory layout. Refer to ASCII table on canvas under handouts. Find hex of character 'A']

Task 1: %n

- **Step 6:** Enter “next” one more time and then again observe the memory layout:

  next

  p &i

  x/16xb &i

- **Step 7:** Exit from gdb:

  exit

- Answer questions on paper after completing all steps

Task 2: %hn

- Change %n to %hn in the printf line
- Repeat steps 2 to 7 from task 1
- Answer questions on paper after completing all steps

Task 3 [Optional]: %hhn

- Change %hn to %hhn in the printf line
- Repeat steps 2 to 7 from task 1